

Page Funnel Visits Analysis

Instituto Tecnológico y de Estudios Superiores de Occidente



Alumno: Luis Fernando Márquez Bañuelos

Profesor: Rodolfo Slay

Materia: Modelos de Crédito



If a step is a question, answer the question after you solve it! ->
 print("This is the answer")

```
In [1]: import pandas as pd
import plotly.graph_objects as go
from IPython.display import display
```

Import all the files

```
In [2]: visits = pd.read_csv('data/visits.csv',
                             parse_dates=[1])

cart = pd.read_csv('data/cart.csv',
                   parse_dates=[1])

checkout = pd.read_csv('data/checkout.csv',
                       parse_dates=[1])

purchase = pd.read_csv('data/purchase.csv',
                       parse_dates=[1])
```

Step 1: Inspect the DataFrames using `print` and `head`

```
In [3]: display(visits.head())
display(cart.head())
display(checkout.head())
display(purchase.head())
```

	user_id	visit_time
0	943647ef-3682-4750-a2e1-918ba6f16188	2017-04-07 15:14:00
1	0c3a3dd0-fb64-4eac-bf84-ba069ce409f2	2017-01-26 14:24:00
2	6e0b2d60-4027-4d9a-babd-0e7d40859fb1	2017-08-20 08:23:00
3	6879527e-c5a6-4d14-b2da-50b85212b0ab	2017-11-04 18:15:00
4	a84327ff-5daa-4ba1-b789-d5b4caf81e96	2017-02-27 11:25:00

	user_id	cart_time
0	2be90e7c-9cca-44e0-bcc5-124b945ff168	2017-11-07 20:45:00
1	4397f73f-1da3-4ab3-91af-762792e25973	2017-05-27 01:35:00
2	a9db3d4b-0a0a-4398-a55a-ebb2c7adf663	2017-03-04 10:38:00
3	b594862a-36c5-47d5-b818-6e9512b939b3	2017-09-27 08:22:00
4	a68a16e2-94f0-4ce8-8ce3-784af0bbb974	2017-07-26 15:48:00

	user_id	checkout_time
0	d33bdc47-4afa-45bc-b4e4-dbe948e34c0d	2017-06-25 09:29:00
1	4ac186f0-9954-4fea-8a27-c081e428e34e	2017-04-07 20:11:00
2	3c9c78a7-124a-4b77-8d2e-e1926e011e7d	2017-07-13 11:38:00
3	89fe330a-8966-4756-8f7c-3bdbcd47279a	2017-04-20 16:15:00
4	3ccdaf69-2d30-40de-b083-51372881aedd	2017-01-08 20:52:00

	user_id	purchase_time
0	4b44ace4-2721-47a0-b24b-15fbfa2abf85	2017-05-11 04:25:00
1	02e684ae-a448-408f-a9ff-dcb4a5c99aac	2017-09-05 08:45:00
2	4b4bc391-749e-4b90-ab8f-4f6e3c84d6dc	2017-11-20 20:49:00
3	a5dbb25f-3c36-4103-9030-9f7c6241cd8d	2017-01-22 15:18:00
4	46a3186d-7f5a-4ab9-87af-84d05bfd4867	2017-06-11 11:32:00

Step 2: Left merging visits and cart

```
In [4]: visits_cart = visits.merge(cart, how='left')
visits_cart
```

Out [4]:

	user_id	visit_time	cart_time
0	943647ef-3682-4750-a2e1-918ba6f16188	2017-04-07 15:14:00	NaT
1	0c3a3dd0-fb64-4eac-bf84-ba069ce409f2	2017-01-26 14:24:00	2017-01-26 14:44:00
2	6e0b2d60-4027-4d9a-babd-0e7d40859fb1	2017-08-20 08:23:00	2017-08-20 08:31:00
3	6879527e-c5a6-4d14-b2da-50b85212b0ab	2017-11-04 18:15:00	NaT
4	a84327ff-5daa-4ba1-b789-d5b4caf81e96	2017-02-27 11:25:00	NaT
...	...	...	...
1995	33913ac2-03da-45ae-8fc3-fea39df827c6	2017-03-25 03:29:00	NaT
1996	4f850132-b99d-4623-80e6-6e61d003577e	2017-01-08 09:57:00	NaT
1997	f0830b9b-1f5c-4e74-b63d-3f847cc6ce70	2017-09-07 12:56:00	NaT
1998	b01bffa7-63ba-4cd3-9d93-eb1477c23831	2017-07-20 04:37:00	NaT
1999	0336ca81-8d68-443f-9248-ac0b8ad147d5	2017-11-15 10:11:00	NaT

2000 rows x 3 columns

Step 3: How long is `visits_cart` ?

```
In [5]: print("This is the answer:")
        print(f'visits_car has a length of: {len(visits_cart)}')
```

This is the answer:
visits_car has a length of: 2000

Step 4: How many timestamps are null for `cart_time` ?

```
In [6]: null_cart_time = visits_cart.cart_time.isnull().sum()

        print("This is the answer:")
        print(f'The number of null cart_time values is: {null_cart_time}')
```

This is the answer:
The number of null cart_time values is: 1652

Step 5: What percentage only visited?

```
In [7]: visited_not_cart = null_cart_time / len(visits_cart)
```

```
print("This is the answer:")  
print(f'The percentage of visits that did not add to cart is: {visited_not_c
```

This is the answer:

The percentage of visits that did not add to cart is: 82.60%

Step 6: What percentage placed a t-shirt in their cart but did not checkout?

```
In [8]: cart_checkout = cart.merge(checkout, how='left')  
  
not_checkout = cart_checkout.checkout_time.isnull().mean()  
  
print("This is the answer:")  
print(f"{not_checkout:.2%} of shoppers placed a t-shirt in their cart but di
```

This is the answer:

35.06% of shoppers placed a t-shirt in their cart but did not checkout.

Step 7: Merge it all together

```
In [9]: all_data = visits_cart.merge(checkout, how='left').merge(purchase, how='left')  
all_data
```

Out [9]:

	user_id	visit_time	cart_time	checkout_time	purchase_time
0	943647ef-3682-4750-a2e1-918ba6f16188	2017-04-07 15:14:00	NaT	NaT	NaT
1	0c3a3dd0-fb64-4eac-bf84-ba069ce409f2	2017-01-26 14:24:00	2017-01-26 14:44:00	2017-01-26 14:54:00	2017-01-26 15:08:00
2	6e0b2d60-4027-4d9a-babd-0e7d40859fb1	2017-08-20 08:23:00	2017-08-20 08:31:00	NaT	NaT
3	6879527e-c5a6-4d14-b2da-50b85212b0ab	2017-11-04 18:15:00	NaT	NaT	NaT
4	a84327ff-5daa-4ba1-b789-d5b4caf81e96	2017-02-27 11:25:00	NaT	NaT	NaT
...	...	...	...	...	...
2103	33913ac2-03da-45ae-8fc3-fea39df827c6	2017-03-25 03:29:00	NaT	NaT	NaT
2104	4f850132-b99d-4623-80e6-6e61d003577e	2017-01-08 09:57:00	NaT	NaT	NaT
2105	f0830b9b-1f5c-4e74-b63d-3f847cc6ce70	2017-09-07 12:56:00	NaT	NaT	NaT
2106	b01bffa7-63ba-4cd3-9d93-eb1477c23831	2017-07-20 04:37:00	NaT	NaT	NaT
2107	0336ca81-8d68-443f-9248-ac0b8ad147d5	2017-11-15 10:11:00	NaT	NaT	NaT

2108 rows x 5 columns

Step 8: % of users who got to checkout but did not purchase

```
In [10]: purchased_checkout = all_data.purchase_time.notnull().sum() / all_data.checkout_time.notnull().sum()
print("This is the answer:")
print(f'The percentage of shoppers that checked out and not purchased is: {100 * purchased_checkout}%')
```

This is the answer:

The percentage of shoppers that checked out and not purchased is: 24.55%

Step 9: check each part of the funnel, let's print all 3 of them again

```
In [11]: print("This is the answer:")  
print("{} percent of users who visited the page did not add a t-shirt to the  
print("{} percent of users who added a t-shirt to their cart did not checkou  
print("{} percent of users who made it to checkout did not purchase a shirt"
```

This is the answer:

82.6 percent of users who visited the page did not add a t-shirt to their cart

35.06 percent of users who added a t-shirt to their cart did not checkout

24.55 percent of users who made it to checkout did not purchase a shirt

The weakest part of the funnel is clearly getting a person who visited the site to add a tshirt to their cart. Once they've added a t-shirt to their cart it is fairly likely they end up purchasing it. A suggestion could be to make the add-to-cart button more prominent on the front page.

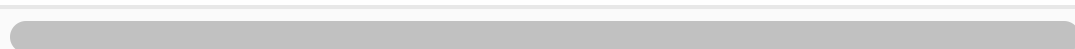
Step 10: adding new column

```
In [12]: all_data['time_to_purchase'] = all_data.purchase_time - all_data.visit_time  
all_data
```

Out [12]:

	user_id	visit_time	cart_time	checkout_time	purchase_time	time_to_pi
0	943647ef-3682-4750-a2e1-918ba6f16188	2017-04-07 15:14:00	NaT	NaT	NaT	
1	0c3a3dd0-fb64-4eac-bf84-ba069ce409f2	2017-01-26 14:24:00	2017-01-26 14:44:00	2017-01-26 14:54:00	2017-01-26 15:08:00	0 days 0
2	6e0b2d60-4027-4d9a-babd-0e7d40859fb1	2017-08-20 08:23:00	2017-08-20 08:31:00	NaT	NaT	
3	6879527e-c5a6-4d14-b2da-50b85212b0ab	2017-11-04 18:15:00	NaT	NaT	NaT	
4	a84327ff-5daa-4ba1-b789-d5b4caf81e96	2017-02-27 11:25:00	NaT	NaT	NaT	
...	...	...	...	...	...	...
2103	33913ac2-03da-45ae-8fc3-fea39df827c6	2017-03-25 03:29:00	NaT	NaT	NaT	
2104	4f850132-b99d-4623-80e6-6e61d003577e	2017-01-08 09:57:00	NaT	NaT	NaT	
2105	f0830b9b-1f5c-4e74-b63d-3f847cc6ce70	2017-09-07 12:56:00	NaT	NaT	NaT	
2106	b01bffa7-63ba-4cd3-9d93-eb1477c23831	2017-07-20 04:37:00	NaT	NaT	NaT	
2107	0336ca81-8d68-443f-9248-ac0b8ad147d5	2017-11-15 10:11:00	NaT	NaT	NaT	

2108 rows x 6 columns



Step 11: examine the results


```
In [13]: all_data.time_to_purchase.describe()
```

```
Out[13]: count                252
mean      0 days 00:43:12.380952380
std       0 days 00:14:42.168126436
min              0 days 00:09:00
25%              0 days 00:32:00
50%              0 days 00:44:30
75%              0 days 00:54:00
max              0 days 01:17:00
Name: time_to_purchase, dtype: object
```

Step 12: what is the average time to purchase?

```
In [14]: avg_time = all_data.time_to_purchase.mean()

print("This is the answer:")
print(f"The average time from visit to purchase is: {avg_time}")
```

This is the answer:

The average time from visit to purchase is: 0 days 00:43:12.380952380

Sankey Diagram

```
In [15]: # Calculate the flow values for the Sankey diagram
total_visits = len(all_data)
visited_to_cart = all_data.cart_time.notnull().sum()
visited_no_cart = total_visits - visited_to_cart
cart_to_checkout = all_data.checkout_time.notnull().sum()
cart_no_checkout = visited_to_cart - cart_to_checkout
checkout_to_purchase = all_data.purchase_time.notnull().sum()
checkout_no_purchase = cart_to_checkout - checkout_to_purchase

# Create Sankey diagram
fig = go.Figure(data=[go.Sankey(
    node=dict(
        pad=15,
        thickness=20,
        line=dict(color="black", width=0.5),
        label=["Visited Page", "Added to Cart", "Reached Checkout", "Purchased",
              "Cart Not Purchased", "Checkout Not Purchased", "Purchase Not Completed"],
        color=["blue", "green", "orange", "purple", "red", "red", "red"]
    ),
    link=dict(
        source=[0, 0, 1, 1, 2, 2], # indices correspond to labels
        target=[1, 4, 2, 5, 3, 6],
        value=[visited_to_cart, visited_no_cart, cart_to_checkout, cart_no_checkout, checkout_to_purchase, checkout_no_purchase]
    )
)])

fig.update_layout(title_text="E-commerce Funnel - Sankey Diagram", font_size=14)
fig.show()
```

E-commerce Funnel - Sankey Diagram

